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# Millikan Oil Drop Outline

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**SUMMARY**
The elementary electric charge $e$ is a fundamental constant that is used in calculations across physics. Therefore accurately measuring it's value is crucial to every field of physics. By analyzing the free-fall motion of electrically charge oil drops in a constant electric field, we demonstrate that charge is a quantized value and calculate the elementary charge $e$. The charge is derived from simple Newtonian physics, providing an estimate of $e$ based solely on measurable quantities from our simple experimental set up. [Comparison to literature value]. This experiment provided the first estimate of the charge of the electron, a major breakthrough for the field of physics.

## 1 INTRODUCTION

Understanding the properties of fundamental particles is of utmost importance for the fields of physics and chemistry. At the turn of the 20th century, the existence of the electron was widely accepted but little was known about it except its charge to mass ratio $q/m$. Robert Millikan devised a way to separate these quantities in his oil drop experiments which he performed between 1909 and 1913, earning him the Nobel Prize in Physics in 1923 [1]. The Millikan Oil Drop Experiment is a classic physics experiment which produced the first measure of the fundamental electron charge $e$ [2].

The experiment takes advantage of simple Newtonian mechanics and electrodynamics to estimate the elementary electric charge. The idea is to observe the motion of charged drops of oil with the forces from gravity, viscous drag, and an applied electric field acting on them. When the forces are balanced such that the drops move at terminal velocity, this analysis leaves a relatively simple expression for the electron charge in terms of measurable experimental quantities.

Importantly, the Millikan Oil Drop experiment further provides evidence for the quantization of charge, because each microscopic oil drop contains an integer number of electrons. Thus, the charge of the droplets can be shown to fit a step-wise function, indicating that electric charge is not a continuous variable. In this experiment, we aim to calculate the fundamental constant $e$ and demonstrate that it is quantized using the same experimental apparatus as Millikan.

## 2 THEORY

The theory of this experiment is straightforward. By simply analyzing the forces acting on a singular oil drop, we can derive an expression for $q$ in terms of experimental values. The full derivation is included in Appendix A, and the main ideas are presented in this section. A description of the experimental set up is presented in Section 4.2.

The left side of Figure 1 shows the forces present on a single oil drop when no electric field is applied. In this case, gravity pulls the drop down towards the bottom plate of the parallel plate capacitor while the viscous drag and buoyancy forces push up. The force of gravity is described by $F_g = mg$. The viscous drag force, $F_v$ is given by:

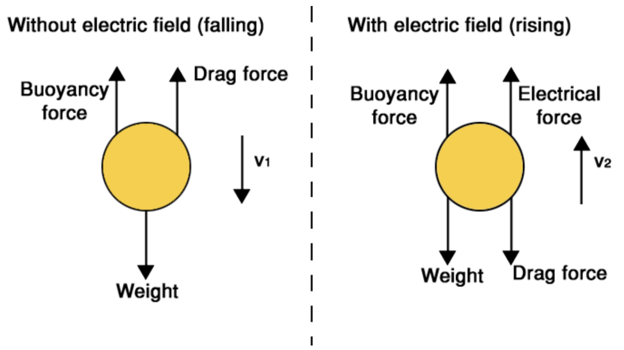


**Figure 1.** Free-body diagrams of an oil drop in free-fall (left) and with an electric field applied (right). Without the presence of the electric field, the drop falls down towards the bottom plate of the capacitor. When there is an electric field, however, the motion of the drop is reversed. Credit: Sam Brind (`https://owlcation.com/stem/Millikans-Oil-Drop-Experiment`).

$$F_v = 6\pi\eta a v_g \tag{1}$$

where $\eta$ is the viscosity of air, $a$ is the radius of the oil drop, and $v_g$ is the velocity the drop has due to gravity. The buoyancy force is given by:

$$F_b = \rho V_{\mathrm{d}} g \tag{2}$$

where $\rho$ is the density of air, $V_{\mathrm{d}}$ is the volume of the drop, or $4/3\pi a^3$, and $g$ is the acceleration due to gravity.

When the electric field is applied inside the parallel plate capacitor, the forces act as shown in the right panel of Figure 1. The field from the parallel plate capacitor is given by $E = Vd$ where $V$ is the voltage applied and $d$ is the separation between the two plates. From Maxwell's equations, the electric force on the drop is therefore $F = qE$ where q is the charge we are interested in. Balancing all four forces yields:

$$qE = mg + 6\pi\eta a v_g \tag{3}$$

The radius of the drop is too small to measure, but because the density is known, we can find the radius from the gravitational and viscous drag forces using $m = \rho V$. Setting $E = 0$ and simplifying yields:

$$a = \left(\frac{9\eta v_g}{2\rho g}\right)^{1/2} \tag{4}$$

Now, using a non-zero field and substituting in the drop radius, we can find a simplified expression for the charge $q$ from experimental variables:

$$q = \frac{6\pi d}{V}\sqrt{\frac{9\eta^3}{2\rho g}}(v_g + v_E)\sqrt{v_g} \tag{5}$$

where $v_E$ is the velocity of the drop in the presence of the electric field. However, the assumptions used to derive the viscous drag force violate assumption in Stoke's Law that the droplets can act as a continuous fluid within the air. Because the radius of the oil drop is not negligible compared to the mean free path of the air molecules it moves through, we must add a correction factor to $\eta$. The final expression for the charge is given by:

$$q = \frac{6\pi d}{V}\sqrt{\frac{9\eta^3}{2\rho g}}(v_g + v_E)\sqrt{v_g}(1 + \frac{b}{aP})^{-3/2} \tag{6}$$

where $P$ is the pressure and $b$ is an empirical constant. Because the correction factor must be unitless, the units of $b$ are length×pressure, and it is commonly report in m(cm Hg).

## 3 SAFETY

Safety is a crucial concern in this lab because of the high voltages required during data taking. The power source for the capacitor usually outputs $\sim 500V$ of electricity. Touching the capacitor while this voltage is applied will result in a painful and potentially dangerous shock. Therefore, we will maintain awareness of when the power source is and is not applied to the capacitor, and remind each other never to touch the apparatus while the voltage is applied. Because we will have to clean oil off of the capacitor, we will always double check that the power supply is off before cleaning.

## 4 EXPERIMENTAL METHODS

In this section we list the materials and apparatus that will be used for the experiment. We then detail our data taking procedure.

### 4.1 Materials

The required materials for this experiment are:

- watch oil
- parallel plate capacitor
- microscope with scale
- camera
- high-voltage power supply

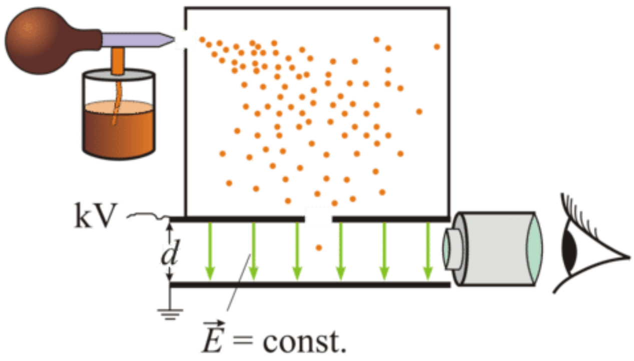


**Figure 2.** Experimental set up of the Millikan Oil Drop experiment. Microscopic oil drops are sprayed from the atomizer into a parallel plate capacitor

- computer for image analysis
- USB cable to connect camera to computer
- lens to focus or enlarge oil drops
- optical rail for alignment
- light source to illuminate drops
- wire to focus microscope

### 4.2 Experimental Set Up

Figure 2 shows our experimental set up. We use a parallel plate capacitor, the black lines, which produce a constant electric field. A microscope is set up to focus on the space in between the plates, and focus a camera through the microscope barrel to improve data tracking. Note that the microscope has a glass scale built in. These elements are aligned linearly on a level plane. We also shine a light at an angle of about 30° to illuminate the drops.

The capacitor is connceted to Matsusada ES Series, model ES-1P3-LSh power-supply generating about 500 V. This produces a constant electric field. We use Nye Watch Oil 140C inside an atomizer, shown in orange. As the drops leave the atomizer, they are charged by friction. They then fall because of gravity through a small slit at the top of the capacitor where they then interact with the electric field.

### 4.3 Experimental Procedure

After aligning the set up, we ensure the camera is focused by dangling the wire in the capacitor where the drops will be. We then remove the wire and carefully turn on the power supply, being mindful not to touch it again until we have taken our data and switched it back off. We begin recording a video with the camera. We use the atomizer to spray oil drops over the parallel plate capacitor.

Once the oil drops have finished moving, we use tracking software to record the position as a function of time of the drops. This involved manually marking the position of the drop in each frame to generate graphs of position and velocity. We complete this analysis with and without an applied field which allows us to solve for the charge given by Equation 6. We will repeat this process as many times as necessary to generate a sufficiently large dataset.

## 5 DATA ANALYSIS AND INTERPRETATION

We will first show that charge is quantized by plotting the results of Equation 6 for multiple oil drops. This will not intrinsically generate the electric charge $e$ because each drop will likely have more than one electron. However, we can show that each drop has a charge that is some integer multiple of $e$ by plotting the charge of several drops. Carefully accounting for the measurement errors will show that the error bars are not large enough to account for the gaps in charge. We can find the average height of each plateau, then take the mean separation between these to generate an estimate for $e$. Once we have this data, we will take a weighted average of the points on each plateau and compare the values to the height of the first plateau to compare them.

Expected sources of uncertainty include measurement uncertainties in every parameter of Equation 6. These will be carefully propagated by the standard formula:

$$\sigma_f^2 = \sum_i \sigma_{a_i}^2 \frac{\partial f}{\partial a_i}^2 \tag{7}$$

where the $a_i$ is each measured quantity in our final expression. We will strive to minimize errors by taking multiple measurements and beginning data analysis early to catch and correct potential systematic errors.

## 6 CONCLUSIONS

In this experiment we reproduced the Millikan Oil Drop experiment to demonstrate that charge is a quantized property of electrons and calculate the fundamental electric charge. Applying simple Newtonian physics allowed us to derive an expression for electron charge from measurable quantities alone. Careful error analysis showed that the gaps between oil drop charges are too large to be attributed to errors alone. We will compare our final value for $e$ to the accepted value in the literature.

[1] Millikan Oil Drop Experiment v1.6, Carmen/Canvas, Aug 2024.

[2] Millikan., R. A., (1913). On the Elementary Electrical Charge and the Avogadro Constant. Phys. Rev. 2(2), 109–143. 10.1103/PhysRev.2.109.

## APPENDIX A: FULL DERIVATION OF CHARGE EQUATION

Will include derivation here.